Venkata Lakshmi Parimala Pasupuleti

Senior Platform Software Engineer · Distributed Systems · Cloud Infrastructure · Identity & Access Management (IAM) · Authentication & Authorization · Microservices · FedRAMP/PCI DSS · Java · Python · Kubernetes · AWS · Observability

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PROFESSIONAL SUMMARY

I build reliable, **fault-tolerant distributed systems** and **microservices** at scale — **Python + Java, Kubernetes, CI/CD, GitOps, observability pipelines** — with 9+ years shipping production services in cloud environments. At Intuit, own **IAM policy enforcement** and **role-based access controls** for financial data services on **AWS** (EC2, RDS, S3, Lambda, **IAM**); full **observability** stack: **OpenTelemetry**, **Prometheus**, **Grafana**, and automated alerting for SLO breach detection; infrastructure automation via **Terraform** IaC, **Kubernetes** orchestration, and **GitHub Actions CI/CD**. At Entrata (100K+ MAU multi-tenant SaaS), implemented **authentication** and **authorization** patterns including **RBAC** and least-privilege access across regulated workflows; HUD/PHI compliance controls aligned to **FedRAMP/PCI DSS** adjacent security frameworks; **HIPAA** data migration with **Libsodium encryption**, **audit trails**, and **incident response** procedures; built large-scale ETL with checkpoint/rollback; containerized **microservices** on **Linux** throughout. Built **HITL governance platform** (LeaseI) with **policy enforcement**, **least-privilege access**, and full **auditability**; **postmortem**-driven reliability improvements. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Identity & Access Management (IAM) · Authentication & Authorization · Authentication Protocols · FedRAMP / PCI DSS Compliance · RBAC & Least-Privilege Access · Microservices Architecture · Service-Oriented Architectures · Incident Response & On-Call · Postmortem Activities · Distributed Systems · Fault-Tolerant Design · Java & Spring Boot · Python (9+ yrs) · CI/CD & GitOps · Observability & Telemetry · OpenTelemetry / Prometheus / Grafana · Kubernetes & Docker · AWS Cloud Infrastructure · Terraform IaC · Linux / Unix Systems · Circuit Breakers & Retry Mechanisms · Policy Enforcement & Auditability · Security & Encryption

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — Cloud Infrastructure · IAM · Authentication & Authorization · Microservices · Observability · AWS · Python · Java

- Own production **microservices** on **AWS** (EC2, RDS, S3, Lambda, API Gateway, **IAM**) for financial data processing; enforce **role-based access controls (RBAC)**, **least-privilege IAM policies**, and **authentication/authorization** patterns across regulated financial data workflows; full **observability** stack: **OpenTelemetry**, **Prometheus**, **Grafana**, and automated **alerting** for SLO breach detection.
- On-call rotation for production financial services: **incident response** triage, root-cause analysis, and **postmortem activities** with corrective action tracking; build Python/FastAPI **REST APIs** with input validation, error handling, **retry mechanisms**, and **circuit breaker** patterns; ship via **GitHub Actions CI/CD** with **GitOps**-style deployment workflows and **rollback and recovery** on failed deploys.
- Implement sensitive-data guardrails and **security controls** for AI-generated outputs integrated into financial applications (Anthropic Claude); **policy enforcement** and **auditability** at every LLM call boundary; LLM-as-judge eval CI prevents quality regressions before production deploy.
- Infrastructure automation: **Terraform** IaC for AWS resource provisioning; **Kubernetes** orchestration of containerized **microservices**; logging and **telemetry** for production visibility across distributed **service-oriented architectures**.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, multi-tenant platform

Senior Software Engineer — Java · Spring Boot · Python · Microservices · IAM · Authentication · Compliance Frameworks · PostgreSQL · Linux · AWS

- Designed and operated **fault-tolerant RabbitMQ**-based **async processing** system for bulk rent updates across thousands of tenant records on a **distributed** multi-tenant SaaS platform: dead-letter queues, exponential backoff retry, **idempotency** guarantees to prevent double-processing, and consumer worker auto-recovery; **incident response** procedures and **postmortem activities** for production reliability.
- Implemented **authentication**, **authorization**, and **RBAC** across regulated affordable housing workflows; HUD/PHI compliance data pipelines aligned to **FedRAMP/PCI DSS**-adjacent security frameworks: **access controls**, **least-privilege** policies, **encryption** (Libsodium), **audit trails**, and **HIPAA/PHI** data protection; loaded 1.6M HUD/Tax Credit records with validation and replication-safe idempotent writes.
- Delivered 20+ production features end-to-end in **Java/Spring Boot** across **service-oriented architectures**; containerized **microservices** with **Docker** (50% scalability improvement, 30% faster deployments); optimized 200+ SQL queries (30% performance gain); built Python/**Bash** automation on **Linux** eliminating 80% of manual operational work.
- **AWS S3** PHI migration with **Libsodium encryption** and HIPAA compliance; Redis **caching** layer for high-volume async workflows; hardened **security** posture across multi-tenant platform with **on-call rotation** support.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Java · Python · PostgreSQL · AWS · Linux

- Built property management modules and compliance data pipelines in **Java**, Python, PHP, and PostgreSQL; delivered large-scale data import/reconciliation workflows (1.6M HUD/Tax Credit records); led **AWS S3** cloud migration from legacy server storage on **Linux** infrastructure; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

ML Research · Fine-Tuning · PyTorch · Statistical Modeling · HuggingFace

- Fine-tuned classification model (HuggingFace Transformers + PyTorch): full pipeline (data prep → fine-tuning → eval → deploy on Quart); Isolation Forest anomaly detection on research datasets; predictive models ($R^2 = 0.977$); presented to GMU faculty panel. Mentored 600+ students.

PORTFOLIO PROJECTS

AgentProbe — Open-Source AI Developer Platform & LLM Evaluation Toolkit

Developer Platform · Agentic Workflows · Observability

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Pinecone, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- **Developer platform** for building and evaluating **AI-assisted developer tools**: 15-endpoint FastAPI backend routing **agentic workflows** across 5 LLM providers; LLM Router for dynamic provider selection based on latency, cost, and capability; **circuit breaker** patterns and fallback routing at every provider boundary; **artifact management** for model evaluation snapshots and golden datasets.
- 5-metric eval framework (factual accuracy, hallucination, retrieval quality, completeness, compliance) with **CI/CD**-integrated regression detection; golden-dataset **test automation** gates deploys on quality drops; **supply-chain security** controls validate provider responses before pipeline consumption; **policy enforcement** and **compliance automation** hooks ensure eval outputs are auditable.

Leasel — Agentic Engineering Platform with Safety Controls & HITL Governance

Agentic Engineering · Safety Controls · Human Oversight

2026 · Python, LangGraph (8-agent), FastAPI, hybrid RAG (TF-IDF + semantic reranking), SQLite, CI/CD · github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Agentic engineering platform** with a **control plane** enforcing **safety controls for autonomous systems**: 8-agent LangGraph workflow (triage → retrieval → parallel agents → reasoning → deterministic validator → confidence-gated **human escalation**); adversarially tested **safety boundary** hard-routes eligibility and financial decisions to human operators, enforcing **human oversight** and **least-privilege access** for high-stakes agentic decisions; **policy enforcement** layer prevents autonomous action without explicit authorization; full **auditability** of every agent decision path.
- LLM-as-judge CI harness: 93.6% precision, 87.9% F1 across 1,240 test cases; deterministic validator enforces **policy-as-code** so regulated decisions stay outside LLM reasoning path; **GitHub Actions CI/CD** gates deploys on quality regression; **rollback and recovery** on eval failure; **workflow orchestration** across 8 agents with explicit state machines.

AI on FHIR — Clinical NLP Backend with HIPAA Security Controls

Backend Services · Security Controls · HIPAA Compliance

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP pipeline translating English/Spanish questions into FHIR **REST API** calls; HIPAA-compliant **authentication**, **authorization**, and **RBAC** for data isolation and **access controls**; **encryption** and PHI-sensitive-data handling meeting **compliance framework** requirements; ~60% reduction in manual clinical lookup time; containerized with **Docker** on Linux.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Languages & Backend: Java, Spring Boot, Python (9+ yrs), FastAPI, Flask, Django, TypeScript, JavaScript, Node.js, PHP, SQL, PL/SQL, Bash, Kotlin, Scala, GraphQL, C#, .NET

Developer Platform & Agentic Engineering: Developer Platform Engineering, Agentic Workflows, AI-Assisted Developer Tools, Safety Controls for Autonomous Systems, Human-in-the-Loop (HITL), Policy Enforcement, Policy-as-Code, Auditability, Control Planes, Workflow Orchestration, Artifact Management, Deployment Workflows, Rollback & Recovery, Supply-Chain Security, Compliance Automation

Distributed Systems & Reliability: Distributed Systems, Fault-Tolerant Design, Redundancy & Replication, Automatic Failover, Circuit Breakers, Retry Mechanisms, Idempotency, Async Processing, RabbitMQ, Kafka, Celery, Event-Driven Architecture

Observability, Reliability & On-Call: OpenTelemetry, Prometheus, Grafana, Alerting, Incident Response, On-Call Rotations, Postmortem Activities, LLM-as-Judge Eval CI, Load Testing

Cloud & Infrastructure: AWS (EC2, S3, RDS, Lambda, SQS, Glue, IAM, API Gateway), Docker, Kubernetes, Terraform IaC, GitHub Actions CI/CD, GitOps, Release Engineering, Linux/Unix, Microservices, Service-Oriented Architectures, REST APIs

Security, IAM & Compliance: Identity & Access Management (IAM), Authentication, Authorization, Authentication Protocols, RBAC, OAuth, Least-Privilege Access, FedRAMP Compliance, PCI DSS Compliance, HIPAA/PHI Compliance, Encryption (Libsodium, AES), Access Controls, Audit Trails, Input Validation, Sensitive-Data Handling

Databases & Storage: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ChromaDB, Pinecone, ETL Pipelines, Query Optimization, Data Modeling

AI & ML: Anthropic Claude, OpenAI, LangGraph, LangChain, RAG Pipelines, Embeddings, PyTorch, scikit-learn, HuggingFace, spaCy, BERT, Prompt Engineering